



Fig. 3: Accessories required for the project

the automatic folder structure after download. Connect the Arduino board to your computer using the USB cable. The green power LED (labelled PWR) on the Arduino Uno board should glow.

Wait for Windows to begin its driver installation

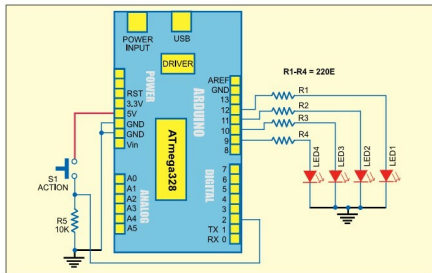


Fig. 4: Circuit of fancy lights controller

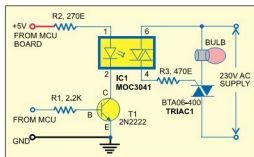


Fig. 5: Circuit of solidstate light switch module

cable or power it with a 9V DC adaptor (or 9V battery) to get started.

Getting started with Arduino (Windows)

Get the latest version of Arduino environment in zip format from the download page <http://arduino.cc/en/Main/Software>. When the download completes, unzip the downloaded file.

Double-click the folder to open it. There should be a few files and subfolders inside. Make sure to maintain

process. After a few moments, the process will fail. Don't worry! Just click 'Start' menu, and open up 'Control Panel'. While in the control panel under 'Category View' option, navigate to 'System and Security'. Next, click 'System'. Once 'System' window is up, open 'Device Manager'. Look under 'Ports' (COM and LPT). You should see an open port named 'Arduino UNO (COMxx)'. Right-click 'Arduino UNO (COMxx)' port and choose 'Update Driver Software' option. Next, choose 'Browse my computer for driver software' option. Finally, navigate to and select the Uno's driver file 'ArduinoUNO.inf', located in 'Drivers' folder of the Arduino software download. This completes the driver installation process of Windows.

Wire the electronic circuit

Follow the wiring diagram shown in Fig. 4 to assemble the lights controller

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

circuit. After construction and physical inspection of the circuit, plug the USB cable of the Arduino into your computer, compile the sketch (Fancy_Lights.pde) and upload it to the Arduino.

Uploading the sketch to Arduino

To upload the sketch, go to your newly unzipped Arduino folder and look for the Arduino IDE icon. Double-click the icon to open the IDE. You will be presented with a blue and white window. Note that before uploading the sketch, you need to tell the IDE which Arduino board and USB port you are using. Go to 'File' menu and click 'Tools,' then click 'Board' option. You will be presented with a list of different Arduino boards that can be connected to the IDE. Choose 'Arduino UNO.' Next, select your USB port. Click 'Tools' again and scroll down to 'Serial Port.' A list of the available serial ports on your system will be displayed. Choose the one that refers to your USB cable.

Now, click 'File' in 'File' menu and select 'New.' Here you can type or copy and paste the code inside the code window. Click 'Verify/Compile' button to check that the code is correct. If everything is fine, upload the code to your Arduino by clicking 'Upload' button. Presuming everything has been set up correctly, you will now see the RX and TX LEDs on the UNO board flash (switch on and switch off) very quickly while data is being uploaded to the MCU in the board. After the data is uploaded to the board successfully, you will get a "Done Uploading" message in the IDE and the RX/TX LEDs will stop flashing. The Arduino will now reset itself and immediately start running the sketch that you have just uploaded.

If everything is okay, remove Arduino from the computer and use it